bias_expert.py — technical

`src/moe_pruning/pruning/bias_expert.py · 46 lines · moe_pruning.pruning.bias_expert`

Purpose

Trainable constant expert.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`class BiasExpert`

lines 17-46 · bases nn.Module

Trainable constant replacement b for a removed expert (gate applied by the parent).

`BiasExpert.__init__`

```
def __init__(self, output_dim: int, init: torch.Tensor | None=None, dtype: torch.dtype | None=None,
device: torch.device | str | None=None)
```

lines 18-33 · raises ValueError

1. Validate `init` shape `(output_dim,)`; copy it (or zeros) to the requested dtype/device; store as `nn.Parameter`.

`BiasExpert.forward`

```
def forward(self, hidden_states: torch.Tensor) -> torch.Tensor
```

lines 35-43

Return b broadcast to `hidden_states.shape[:-1] + (output_dim,)` (un-gated).

1. Output shape = input leading dims + `(output_dim,)`; return `bias.to(input dtype).expand(shape).clone()` - a real copy so downstream in-place ops cannot modify the parameter; gradients flow to `bias`.

Errors raised

Function	Exception	Message (f-string fields in braces)
`__init__`	ValueError	init has shape {tuple(init.shape)}, expected ({output_dim},)